

Statistical Methods

Exam Solution - Moed B

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August 5, 2020

Solution A

1.A.

Proof of variance and bias relationship [\[edit\]](#)

$$\begin{aligned}\text{MSE}(\hat{\theta}) &= \mathbb{E}_{\theta} \left[(\hat{\theta} - \theta)^2 \right] \\ &= \mathbb{E}_{\theta} \left[\left(\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}] + \mathbb{E}_{\theta}[\hat{\theta}] - \theta \right)^2 \right] \\ &= \mathbb{E}_{\theta} \left[\left(\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}] \right)^2 + 2 \left(\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}] \right) \left(\mathbb{E}_{\theta}[\hat{\theta}] - \theta \right) + \left(\mathbb{E}_{\theta}[\hat{\theta}] - \theta \right)^2 \right] \\ &= \mathbb{E}_{\theta} \left[\left(\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}] \right)^2 \right] + \mathbb{E}_{\theta} \left[2 \left(\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}] \right) \left(\mathbb{E}_{\theta}[\hat{\theta}] - \theta \right) \right] + \mathbb{E}_{\theta} \left[\left(\mathbb{E}_{\theta}[\hat{\theta}] - \theta \right)^2 \right] \\ &= \mathbb{E}_{\theta} \left[\left(\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}] \right)^2 \right] + 2 \left(\mathbb{E}_{\theta}[\hat{\theta}] - \theta \right) \mathbb{E}_{\theta} \left[\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}] \right] + \left(\mathbb{E}_{\theta}[\hat{\theta}] - \theta \right)^2 & \mathbb{E}_{\theta}[\hat{\theta}] - \theta = \text{const.} \\ &= \mathbb{E}_{\theta} \left[\left(\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}] \right)^2 \right] + 2 \left(\mathbb{E}_{\theta}[\hat{\theta}] - \theta \right) \left(\mathbb{E}_{\theta}[\hat{\theta}] - \mathbb{E}_{\theta}[\hat{\theta}] \right) + \left(\mathbb{E}_{\theta}[\hat{\theta}] - \theta \right)^2 & \mathbb{E}_{\theta}[\hat{\theta}] = \text{const.} \\ &= \mathbb{E}_{\theta} \left[\left(\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}] \right)^2 \right] + \left(\mathbb{E}_{\theta}[\hat{\theta}] - \theta \right)^2\end{aligned}$$

source: [wikipedia](#)

1.B.

$$y_n \sim N(\mu, \sigma^2)$$

where we are given $\mu = 0$.

$$f(y_n; \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_n - \mu)^2}{2\sigma^2}}$$

iid observations

$$f(y; \sigma^2) = \frac{1}{(2\pi)^{\frac{N}{2}} \sigma^N} e^{-\frac{\sum_{n=1}^N (y_n - \mu)^2}{2\sigma^2}}$$

$$\log f(y; \sigma^2) = C - \frac{N}{2} \log \sigma^2 - \frac{\sum_{n=1}^N (y_n - \mu)^2}{2\sigma^2}$$

$$\max_{\mu, \sigma^2 \geq 0} -\frac{N}{2} \log \sigma^2 - \frac{\sum_{n=1}^N (y_n - \mu)^2}{2\sigma^2}$$

The likelihood equations

$$\frac{\partial f}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \frac{\sum_{n=1}^N (y_n - \mu)^2}{2\sigma^4} = 0$$

Plugging $\mu = 0$ yields

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{n=1}^N y_n^2 \quad \text{automatically non-negative}$$

1.C

$$\begin{aligned} E[\hat{\sigma}^2] &= \frac{1}{N} E \left[\sum_{n=1}^N y_n^2 \right] = \frac{1}{N} N E[y_n^2] = \sigma^2 \Rightarrow Bias = \hat{\sigma}^2 - \sigma^2 = 0 \\ E \left[\frac{1}{N} \sum_{n=1}^N y_n^2 - \sigma^2 \right]^2 &= E \left[\frac{1}{N^2} \sum_{n=1}^N \sum_{m=1}^N y_n^2 y_m^2 - \frac{2\sigma^2}{N} \sum_{n=1}^N x_i^2 + \sigma^4 \right] \\ &= \frac{1}{N^2} \sum_{n=1}^N E[y_n^4] + \frac{1}{N^2} \sum_{n=1}^N \sum_{m \neq n}^N E[y_n^2] E[y_m^2] - \frac{2N\sigma^4}{N} + \sigma^4 \\ &= \frac{1}{N^2} \sum_{n=1}^N 3\sigma^4 + \frac{N^2 - N}{N^2} \sigma^4 - 2\sigma^4 + \sigma^4 = \left(\frac{3}{N} + \frac{N-1}{N} - 2 + 1 \right) \sigma^4 = \frac{2\sigma^4}{N} \end{aligned}$$

Solution B

1. For a valid PDF, we require integration to 1:

$$1 = \int_{-\infty}^{\infty} f(x)dx = \int_0^1 a(1-x)dx = \frac{a}{2} \Rightarrow a = 2$$

2. Calculate the joint distribution using Bayes.

$$\begin{aligned} \bullet f_{Y|X}(y|x) &= \begin{cases} \frac{1}{1-x} & \text{for } x \leq y \leq 1 \\ 0 & \text{else} \end{cases} \\ \bullet f_X(x) &= \begin{cases} 2(1-x) & \text{for } 0 \leq x \leq 1 \\ 0 & \text{else} \end{cases} \end{aligned}$$

Therefore, for $0 \leq x \leq y \leq 1$:

$$f_{X,Y}(x,y) = f_{Y|X}(y|x)f_X(x) = \frac{2(1-x)}{1-x} = 2$$

And the marginal distribution:

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}dx = \int_0^y 2dx = 2y$$

3. MMSE:

$$\hat{X}(Y) = E[X|Y]$$

We need the conditional distribution $X|Y$:

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)f_X(x)}{f_Y(y)} = \frac{2}{2y} = \frac{1}{y}$$

That is, given Y , X is uniformly distributed in $[0, y]$. As a result, the MMSE, which is the conditional expectation is $\hat{X}(Y) = E[X|Y] = \frac{Y}{2}$.

4. We will use previous result: $Y_1|X_1 = \text{uniform}(X_1, 1)$:

$$P(Y_1 > Y_2|X_1 = x_1, X_2 = x_2, Y_2 = y_2) = P(Y_1 \in (y_2, 1)|X_1 = x_1) = \frac{1-y_2}{1-x_1}$$

Now, using the law of total probability we can integrate for all possible Y_2 :

$$\begin{aligned} P(Y_1 > Y_2|X_1 = x_1, X_2 = x_2) &= \int_{x_2}^1 f_Y(y_2|x_1, x_2) \frac{1-y_2}{1-x_1} dy_2 \\ &= \int_{x_2}^{x_1} \frac{1}{1-x_2} dy_2 + \int_{x_1}^1 \frac{1}{1-x_2} \frac{1-y_2}{1-x_1} dy_2 \\ &= \frac{x_1-x_2}{1-x_2} + \frac{y_2-0.5y_2^2}{(1-x_1)(1-x_2)} \Big|_{x_1}^1 \\ &= \frac{0.5(1+x_1)-x_2}{1-x_2} \end{aligned}$$

Solution C

$$1. E[X_n] = \alpha E[X_{n-1}] = \dots = \alpha E[X_0] = 0$$

$$\text{var}(X_n) = \text{var}(\alpha X_{n-1} + W_n) = \alpha^2 \text{var}(X_{n-1}) + \sigma^2$$

To get a stationary process we need $\text{var}(X_n) = \text{var}(X_{n-1}) = \sigma_0^2$:

$$\sigma_0^2 = \alpha^2 \sigma_0^2 + \sigma^2 \Rightarrow \sigma_0^2 = \frac{\sigma^2}{1 - \alpha^2}$$

2.

$$E[X_n X_{n+k}] = E[X_n (\alpha X_{n+k-1} + W_n)] = \alpha E[X_n X_{n+k-1}] + E[X_n W_n]$$

Using the independency $E[X_n W_n] = E[X_n] E[W_n] = 0$. Therefore:

$$E[X_n X_{n+k}] = \alpha E[X_n X_{n+k-1}] = \alpha^2 E[X_n X_{n+k-2}] = \dots = \alpha^k E[X_n X_n] = \alpha^k \sigma_0^2$$

Hence, the process is indeed stationary.

3. we denote $\mathbf{x}_i = \{x_0, x_1, \dots, x_i\}$.

Start with the PDF:

$$f(\mathbf{x}_n) = f(x_n | \mathbf{x}_{n-1}) f(\mathbf{x}_{n-1}) = f(x_n | \mathbf{x}_{n-1}) f(x_{n-1} | \mathbf{x}_{n-2}) f(\mathbf{x}_{n-2}) = \dots = \prod_{i=0}^n f(x_i | \mathbf{x}_{i-1})$$

Now, given X_{i-1} , X_i is independent of all previous observations:

$$f(x_i | \mathbf{x}_{i-1}) = f(x_i | x_{i-1}) \sim N(\alpha x_{i-1}, \sigma^2)$$

$$\text{or: } f(x_i | \mathbf{x}_{i-1}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \alpha x_{i-1})^2}{2\sigma^2}\right)$$

4. **The likelihood:**

$$f(\mathbf{x}_n; \alpha, \sigma^2) = \prod_{i=0}^n f(x_i | \mathbf{x}_{i-1}) = \prod_{i=0}^n f(x_i | x_{i-1}) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{\sum_{i=0}^n (x_i - \alpha x_{i-1})^2}{2\sigma^2}\right)$$

The log-likelihood:

$$LL(\alpha, \sigma^2; \mathbf{x}_n) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{\sum_{i=0}^n (x_i - \alpha x_{i-1})^2}{2\sigma^2}$$

Partial derivative to find α :

$$\frac{\partial}{\partial \alpha} LL = 0 \iff \sum_{i=0}^n (x_i - \alpha x_{i-1}) x_{i-1} = 0$$

$$\text{Solving for } \alpha \text{ results in } \hat{\alpha} = \frac{\sum X_i X_{i-1}}{\sum X_i^2}.$$

5. Since the maximum likelihood estimator for α is independent of σ^2 , we can now assume $\alpha = \hat{\alpha}$ and solve MLE for σ^2 , as following.

Partial derivative to find σ^2

$$\frac{\partial}{\partial \sigma^2} LL = -\frac{n}{2} \frac{1}{\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=0}^n (x_i - \alpha x_{i-1})^2$$

Equating to zero and solving to find σ^2 :

$$\sigma^2 = \frac{\sum_{i=0}^n (x_i - \alpha x_{i-1})^2}{n}$$

In words, we get that the MLE for the variance is the average of the sample variance (over the past observations).